

Permissionless Perps for Any Token

Technical design: mechanism, math, and how the open safety problems get resolved

Grounded in the live engine. Draft for technical feedback.

The goal: let any project open a tradable perp on its token with a small, refundable creator stake instead of deep liquidity or paid market makers. An automated curve is the counterparty from block one; a creator first-loss buffer is the real money behind the trades; the market grows from its own fees.

The safety claim, precisely:

For an isolated market, the protocol and non-creator capital take no loss, and the most that can be extracted is bounded by the market's own capital C_m , with the creator's stake consumed first. Capacity is bounded by:

$$N_{\max} = \beta \cdot C_m / R_{\max}$$

C_m = creator stake + locked fees + extra backing. $\beta < 1$ safety margin. $R_{\max}$ = max loss per position.

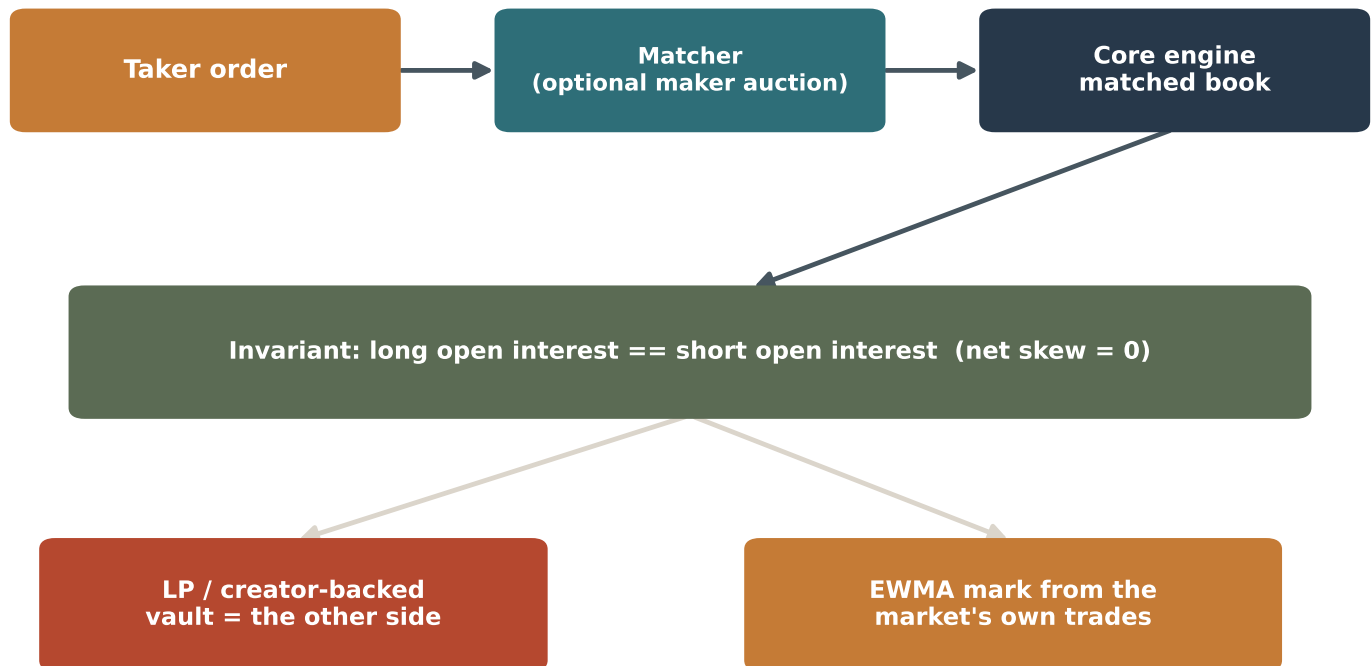
What's true today vs what needs building:

- **The fee / pricing engine is sound now.**

The vol-adaptive curve and matched-book settlement work on the live engine.

- **The capacity bound is not yet enforced in the core engine.**

The four sections that follow turn each open assumption into a concrete, code-level fix with a known residual.



The book is always net-flat: every long is matched by a short. The creator-backed vault account IS the counterparty that fills the gap, so no external market maker is needed. For a token with no oracle, the mark is an EWMA of the market's own trades. Anyone can crank the market permissionlessly to keep it current. This is the foundation the design rests on; it is real and live. What it lacks - a hard capacity cap, a locked buffer, segregated insurance, an ordered loss waterfall - is what the next sections add.

A trade moves the quoted price along a constant-product curve; the fee adapts to volatility; and the size that can be opened is bounded by the buffer. The three pieces:

- **Pricing curve.**

Integer constant-product quoting around the EWMA mark, with a per-slot price clamp so no single slot can move the mark more than a set fraction.

- **Adaptive fee.**

Fee rises with measured volatility (next page). Cheap when calm to attract flow, expensive when wild to protect the buffer and to make manipulation uneconomic.

- **Capacity cap.**

Total open interest is capped so the worst-case loss the market can owe stays inside the buffer. This is the load-bearing safety bound:

$$\text{open interest} \leq N_{\text{max}} = \beta \cdot C_m / R_{\text{max}}$$

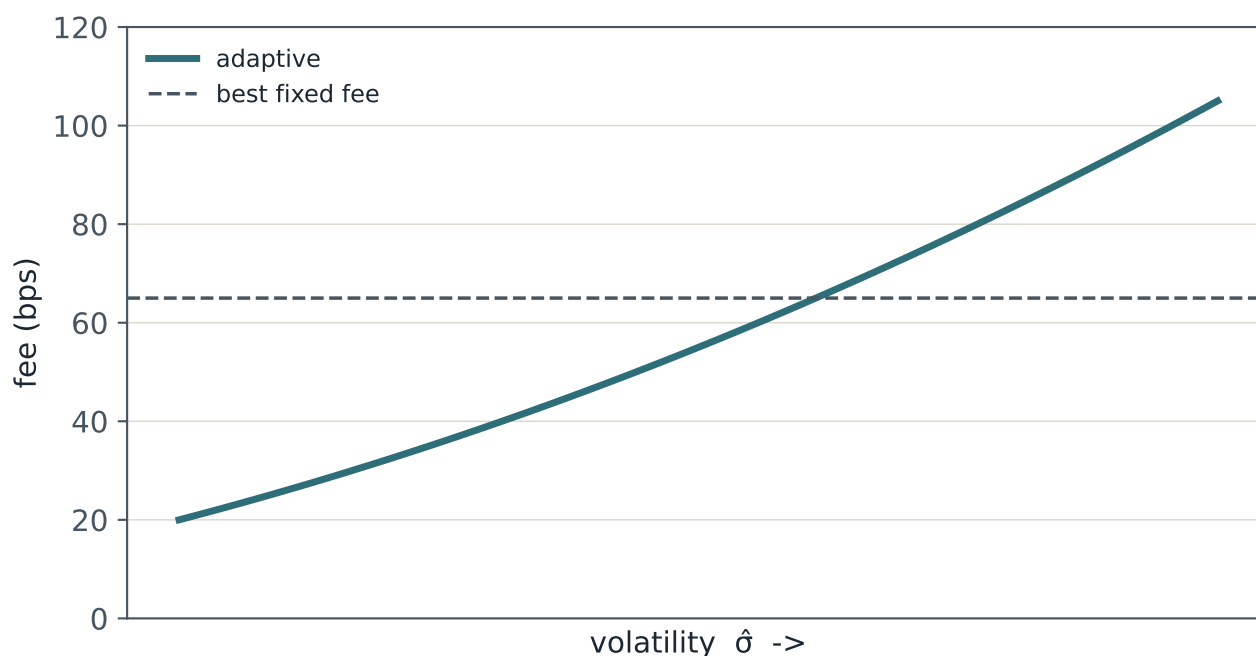
Enforced as a hard reject at trade preflight (proposed; see Fix 1). Today this is unenforced in the core engine.

Everything downstream - the loss waterfall, the liquidation budget, the manipulation cost - is sized against this same N_{max} . If the cap holds, the rest of the safety argument is a matter of ordering and timing, which the fixes address.

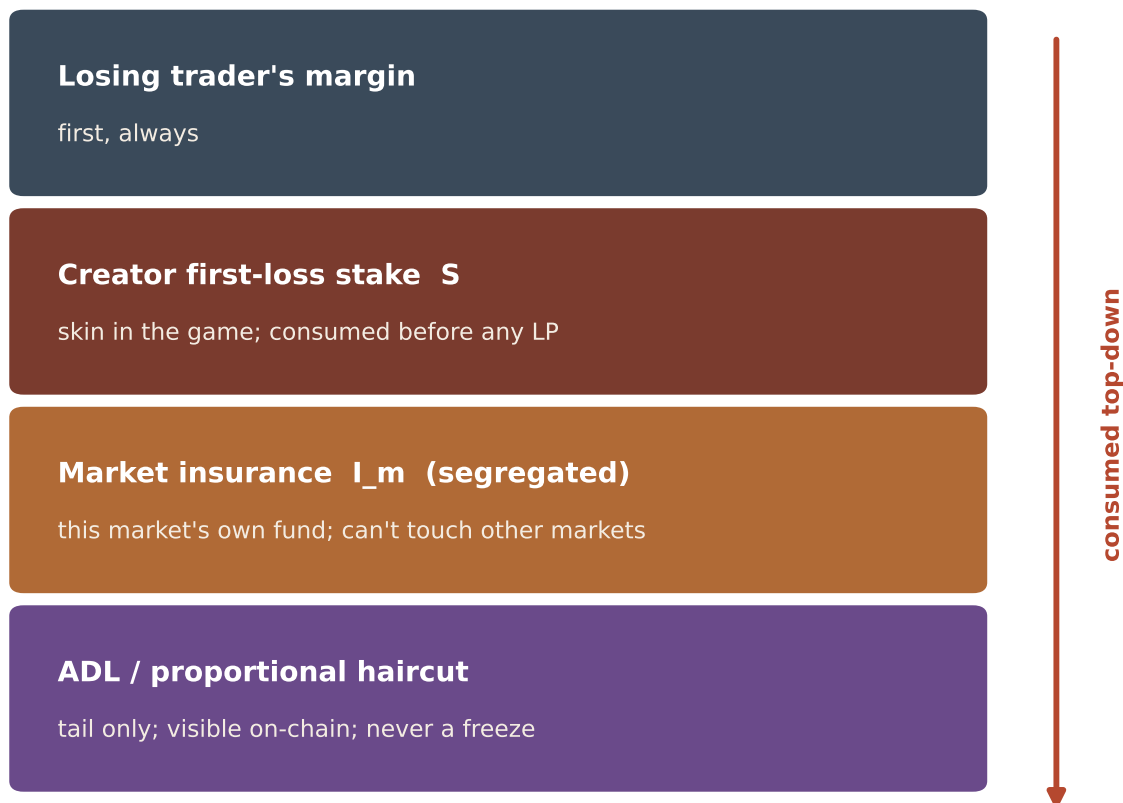
The fee tracks a running estimate of the token's volatility. In a simple, integer-friendly form:

$$\text{fee_bps} = \text{clamp}(a + b \cdot \hat{\sigma} + c \cdot \hat{\sigma}^2, \text{floor}, \text{ceiling})$$

$\hat{\sigma}$ = running volatility estimate (cumulative-mean variance). Calm → near the floor; wild → toward the ceiling.



In simulation across many markets, this adaptive rule beat the best single fixed fee by roughly 6% on captured spread while staying competitive, with bit-for-bit identical behaviour in native and on-chain execution. It charges more only where the risk actually is, which is also what makes the mark expensive to manipulate (Fix 3).



The promise to traders: in normal conditions winners are paid in full by the losing side. Only when claims exceed the whole stack does the bottom layer engage, as a proportional, on-chain-readable haircut - not a freeze, never zero. The key correctness requirement is ORDER: the creator's stake must be spent before any non-creator LP is touched. On the live engine today, the bottom layer (ADL) can fire before the creator layer is exhausted. Fix 1b reorders this.

Problem. Gross OI is unbounded in the core engine (the net-flat invariant bounds skew, not size). And backing can be withdrawn while positions are open, because the withdrawal gate keys on converted PnL, not open OI - so a bond posted to inflate the cap can be pulled right back, dumping the loss on everyone else.

Fix.

```
// at trade preflight
if cap != 0 && open_interest + size > cap { reject }
// at backing withdrawal
require reserved_after ≥ reserved · side_OI / cap
```

$$\text{withdrawable} = \text{reserved} \cdot (1 - \text{side_OI} / \text{N_max})$$

You can only ever pull the un-encumbered fraction. At full OI, nothing; as positions close, it frees up.

Adversarial re-check & residual.

Re-running the self-funding attack: the bond can no longer be withdrawn while it backs open OI - rejected at the floor. Two residuals, both handled: the cap must be tied to current backing (not settable to infinity), and a same-block withdraw-then-open race is itself bounded by the OI cap. Cost: ~200 lines, one new field per market (layout migration), a handful of machine-checkable invariants.

Problem. On bankruptcy, the social-loss step adjusts the opposite side's index, writing losses onto non-creator LPs before the creator stake is used up. And insurance is one shared pool per group, so one market can drain another.

Fix.

Insert a creator-funded layer that is consumed BETWEEN insurance and ADL, so ADL only fires once the creator stake hits zero. Replace the shared pool with per-market segregated insurance held in a per-market vault PDA, so a loss in market A cannot reduce market B's fund.

BEFORE

S=\$50, I_m=\$100, LP=\$500

3 bankrupt longs (\$300)

ADL hits LP early

LP loses \$200

AFTER (creator-first)

S=\$50, I_m=\$100, LP=\$500

insurance, then S, then ADL

creator \$50 absorbed first

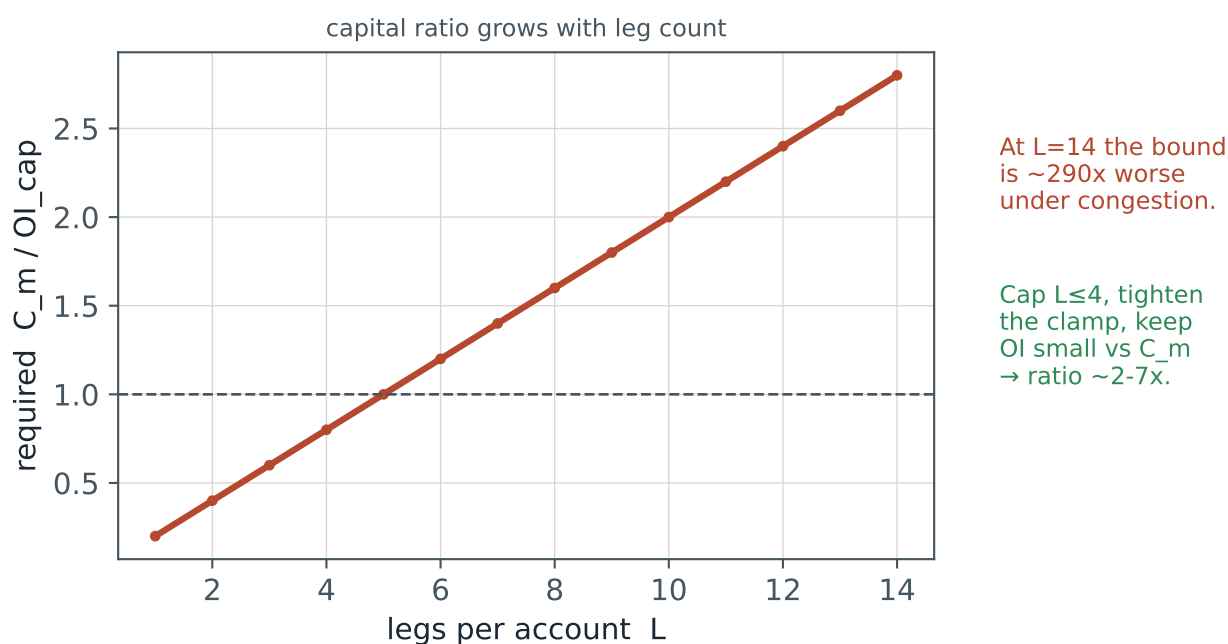
LP loses \$60

Residual: in the deepest tail (creator stake AND insurance both exhausted) ADL still reaches LPs - by design, and only there. The creator stake must be OI-locked (same lock as Fix 1a) so it can't be pulled before it's needed. Cost: ~4 new fields/market, a new vault PDA, a one-time migration.

Close-out is one account, one leg, per transaction; a price-moving crank invalidates other accounts' health certs; a too-fast move halts the market into a safe state. If close-out lags, the per-position loss grows. The honest bound, with a correctly-batched keeper:

$$C_m \geq OI_{cap} \cdot P \cdot 2L / k$$

P = per-slot price clamp, L = max legs per account, k = fraction of slots the keeper lands a tx.



So this caveat is closed mostly by CONFIG, not code: cap legs at 4, set a conservative per-slot clamp, size OI small relative to the buffer, and run the keeper one-leg-per-account in parallel. Under extreme congestion or censorship the market HALTS safely (no new loss accrues) rather than staying open. The price of this safety is capital efficiency: you hold a few times more buffer than open interest.

Internal (own-market).

A depth gate keeps size capped until the market clears a minimum age and volume; a longer EWMA half-life and tight per-slot clamp slow the mark; the OI cap bounds the prize. Together the cost to move the mark exceeds the capped gain:

$$\text{cost to move mark} > \text{max extractable} \leq C_m$$

N_max must be PROTOCOL-enforced, not creator-set - else a creator-as-manipulator breaks even.

External (other protocols).

Zero out the public price field in endogenous-mark mode (sentinel), set a discriminator flag, and warn in the IDL. This stops NAIVE misuse cold. It does NOT stop a determined third party from reading the public mark directly - nothing on-chain can. We classify external use as a documented integrator-discipline risk and state it plainly.

Residual.

A time gate (e.g. several hours) is the hardest control - a wash-trading ring can pay fees to inflate volume but cannot skip the clock. Cost: ~110 lines, a few per-market config knobs, no change to the matched-book core.

For markets that want to promise every winner a full payout no matter what, the haircut tail has to be removed - which means the maximum payout is escrowed up front. The payoff per position is capped, and the reserve covers the whole cap:

Reserve = $OI_cap \cdot (\text{payout cap multiple})$ escrowed before trading

With the max owed pre-funded, the ADL / haircut path is never reached. This is the only mode that is fully safe today.



The trade-off is explicit: it costs real capital and caps the upside on any one position, so it suits tokens or venues that already have funds to reserve and want a bullet-proof guarantee. The cheap, capped-haircut version (Fixes 1-3) is for the long-tail case where pre-funding isn't available; the pre-funded version is the opt-in upgrade for those who can afford certainty.

- **Capital efficiency.**

Safety against lagging liquidation means holding a few times more buffer than open interest. Long-tail markets stay small until they prove out.

- **Deepest tail still reaches LPs.**

Once creator stake and segregated insurance are both exhausted, ADL touches LPs - by design. Only the pre-funded mode removes this entirely.

- **Authority / key management.**

The OI cap and creator-stake controls are only as safe as the keys that set them. Standard multisig / governance discipline applies.

- **Determined external mark misuse.**

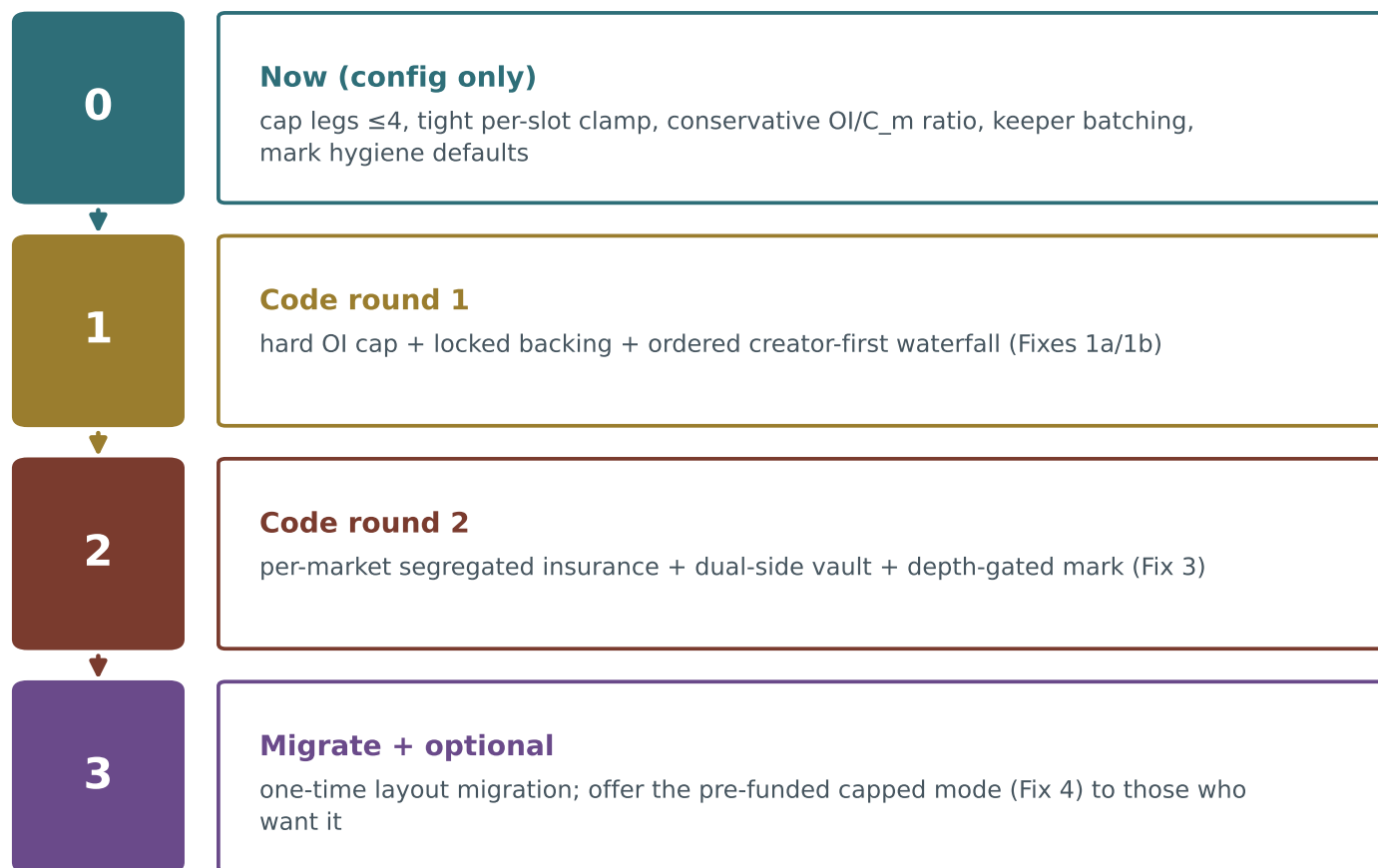
Hygiene stops accidents, not a determined integrator reading the public mark. This cannot be fixed on-chain; it is documented, not eliminated.

- **Congestion / censorship.**

Under sustained censorship the market halts safely rather than staying open. Liveness, not solvency, is what degrades.

- **Migration.**

Existing markets need a one-time, frozen-state migration to the segregated-insurance layout. New markets get it from launch.



Rough size of the coded work: a few hundred lines across the engine and program, four to six new per-market fields, two new instructions, one migration, and a set of machine-checkable invariants for the cap, the waterfall order, and the segregation. None of it touches the matched-book core's settlement math.

- **Capital ratio.**

Is a 2-7x buffer-to-OI ratio acceptable for a long-tail launch, or does that kill the use case versus a smaller, capped product?

- **Capped vs uncapped payout.**

For a brand-new token, will traders accept a rare, visible, proportional haircut in exchange for cheap launch, or do they expect the pre-funded guarantee?

- **Liquidation under congestion.**

Is halt-then-recover an acceptable failure mode, and is the keeper-batching assumption realistic on mainnet?

- **Mark trust.**

Is the depth-gate + time-gate enough for the market's own safety, given external misuse stays a documented risk?

- **Scope.**

Which of the four fixes is worth building first, and is anything missing from the safety argument?